Avik Mondal, PhD

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WORK EXPERIENCE

Aug 2026–Present **Ocean Data Scientist/Computational Modeler (Brandeis University)** *Waltham, MA*

- Developing frequency-domain and wavelet scattering **time-series algorithms** for ill-posed spatial inverse problems, extrapolating continuous 4D dynamical regimes (**1 km–15 km** scales) from a sparse, multi-rate **127-sensor** array.
- **Modeling** ocean flows with **physics-based simulations** in Julia to serve as a null hypothesis for data analysis.
- Engineering an end-to-end pipeline for the **9-node, 127-sensor** marine array (OSMOSIS), synchronizing and cleaning **20M+** asynchronous, multi-rate observations (**5-min to 60-min** intervals) over a **365-day** deployment.

Jan 2026–Mar 2026 **Research Quality Analyst, Flypower.io** *Remote*

- Cross-referenced **AI-synthesized zoning and community-sentiment intelligence** against primary sources, correcting inaccuracies across 15 risk reports supporting 20 GW of geothermal and solar permitting decisions.

May 2021–June 2025 **Time Series Analyst and Algorithm Developer (University of Michigan)** *Ann Arbor, MI*

- Developed a **spatiotemporal spectral decomposition and signal-processing** pipeline over 1 TB of coupled ocean-atmosphere simulation output to quantify turbulent energy transfers and tracer variance budgets and to separate physical dynamical signal from numerical dissipation, published in *Physical Review Fluids* (2025).
- Engineered parallelized **Python HPC pipelines** (Xarray, Dask, Zarr, SLURM) to parse, store, and validate 1 TB of gridded model output as reproducible, analysis-ready **NetCDF and Zarr** datasets with provenance tracking.

May 2019–June 2025 **Computational Modeler/Data Analyst (University of Michigan)** *Ann Arbor, MI*

- Engineered a **C++ multiscale simulation with 100,000+ degrees of freedom** and stochastic nonlinear dynamics, used as a modeling framework to interpret experimental data. Manuscript in progress.
- **Supervised 5 undergraduate researchers** while **fine-tuning an existing computer-vision system** to identify features across 1 TB of microscopy imagery, improving detection consistency across 10 distinct experimental conditions.

Sep 2017–June 2019 **Computational Physics Researcher (UC Santa Barbara Worster Fellow)** *Santa Barbara, CA*

- Developed **physics-informed statistical models and graph-based algorithms** on finite-element structural data, delivering a **9% improvement** in bone-disease predictability over clinical benchmarks through rigorous held-out validation. Published in *Physical Review E* (2019).

PORTFOLIO PROJECTS

Dec 2025–Present **Bayesian Ocean Field Reconstruction (Gaussian Process Regression)** *Waltham, MA*

- Independently building Bayesian GPR surrogate to reconstruct continuous subsurface ocean temperature fields from sparse Argo observations, producing **uncertainty-quantified** profiles across a 2M km² Pacific domain and identifying regions where predictive variance grows.

Mar 2026–Present **Climate-Hazard Risk Model for Aquaculture and Coastal Systems** *Waltham, MA*

- Independently building probabilistic risk model in **PyTorch** ingesting weather and ocean reanalysis time-series to predict financial risk to coastal wildlife and aquaculture, producing uncertainty-quantified geospatial risk maps.

TECHNICAL SKILLS

- **Languages and Frameworks:** Proficient: Python (SciPy, Xarray, Dask), C++ (Eigen). Developing: PyTorch, scikit-learn, Julia, SQL
- **Statistical Modeling and Signal Processing:** Bayesian inference, Gaussian process regression, uncertainty quantification, spatiotemporal spectral decomposition, signal-versus-noise decomposition, time series analysis, held-out model evaluation, bootstrapping, Monte Carlo methods
- **Data Engineering and HPC:** large multidimensional datasets (NetCDF, HDF5, Zarr), high-performance computing clusters (SLURM/PBS), Git, Coiled, provenance tracking. Previous Experience: AWS S3/EC2, GCP
- **Research and Communication:** 2 peer-reviewed first-author publications (*Physical Review Fluids*, *Physical Review E*), 4 conference presentations (AGU, APS), 5 undergraduate researchers mentored, technical documentation

EDUCATION

2018–2025 **University of Michigan** | PhD, Physics (GPA: 4.0) *Ann Arbor, MI*

2014–2018 **University of California, Santa Barbara** | BS, Physics (GPA: 3.95) *Santa Barbara, CA*